Patch Tensor-Based Multigraph Embedding Framework for Dimensionality Reduction of Hyperspectral Images

Yang-Jun Deng, Heng-Chao Li, *Senior Member, IEEE*, Xin Song, Yong-Jian Sun, Xiang-Rong Zhang, *Senior Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Graph-based dimensionality reduction (DR) techniques are of great interest in the field of image processing and especially on the analysis of hyperspectral images (HSIs). Considering the characteristics of hyperspectral data, many different types of graphs were designed to describe the structure of HSIs. Generally, the algorithms based on these graphs achieved promising performance. However, most of them only focus on how to improve the measurement of similarity between the data points by a single graph. Specifically, vector-based graph methods fail to capture the spatial information, while tensor-based graph methods assume that the pixels in each patch tensor belong to the same class, which is not exactly correct in practice. To overcome these shortcomings, this article proposes a patch tensor-based multigraph embedding (PTMGE) framework for the DR of HSIs, in which three different types of subgraphs are constructed to comprehensively describe the intrinsic geometrical structures of HSIs. First, a tensor subgraph is constructed to capture the spatial information and local geometrical structure. Second, for each two neighboring patch tensors in the tensor graph, a bipartite graph is designed to characterize the pixel-based relationships between the patch tensors. Then, considering that the diversity of pixels may be existed in each patch tensor, a pixel-based subgraph is built to describe the inner geometrical structures of every patch tensor. Finally, a novel graph fusion strategy is designed to calculate a final similarity matrix for projection learning. Experiments on three real hyperspectral data sets are conducted, and comparison with some state-of-the-art algorithms validated the effectiveness of our proposed PTMGE method.**

***Index Terms*— Classification, dimensionality reduction (DR), graph embedding (GE), hyperspectral images (HSIs), tensor analysis.**

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Y.-J. Deng, H.-C. Li, and X. Song are with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu 611756, China (e-mail: hcli@home.swjtu.edu.cn).

Y.-J. Sun is with the Department of Traum Orthopedics, The Fifth Affiliated Hospital, Southern Medical University, Guangzhou 510900, China.

X.-R. Zhang is with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China.

Q. Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA.

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I. Introduction

HYPERSPECTRAL remote sensing is an information acquisition technology combining imaging technique and spectral technique, which can simultaneously explore 2-D spatial and 1-D spectral information of ground objects. An obtained hyperspectral image (HSI) consists of hundreds of contiguous narrow spectral bands from visible-to-infrared electromagnetic spectrum, containing abundant land-cover information [1]. Therefore, HSIs have drawn great interests in Earth observations and been successfully applied in many fields, such as agricultural management [2], forest inventory [3], and anomaly detection [4]. Nevertheless, the high spectral resolution of HSIs also brings challenges in data processing and analysis [5]. The overwhelming spectral bands contain great redundancy, which may cause the problem of curse of dimensionality when the labeled training samples are insufficient [6]. Moreover, the high spectral dimensionality not only significantly increases the data storage but also leads to a great increase in computational complexity. Consequently, reducing the dimensionality of HSIs without a significant loss of useful information is of great importance for HSI analysis [7]–[9].

In general, dimensionality reduction (DR) for HSIs can be achieved in two essential ways: feature selection and feature extraction [10]. Feature selection is to obtain a smaller subset of the data, which contains significant and informative features [11]. In HSI processing, feature selection is also known as band selection, and some representative band selection methods include information theoretic band selection methods [12], [13], linear regression-based band selection methods [14], [15], and compressive sensing approaches [16], [17]. This article will focus on DR via feature extraction, which aims at generating a data-driven linear or nonlinear transformation to obtain a transformed version of the data [18]. Ideally, the extracted features should preserve the intrinsic information of the data and contain low-redundancy among features [19], [20]. Up to now, numerous feature extraction-based DR methods have been proposed and achieved promising performance [9], [21]. These methods can be generally divided into unsupervised and supervised ones. The most commonly used unsupervised method is principal component analysis (PCA) [22] that aims to maximize the variance of the transformed data. The typical supervised technique is linear

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discriminant analysis (LDA) [23] that aims to find a projection matrix that can maximize class separability. However, both PCA and LDA fail to discover the nonlinear structure of the data.

In recent years, to explore the nonlinear structure of data, manifold learning-inspired DR methods have been widely applied in computer vision [24], [25]. Especially, in image processing, some studies [26], [27] have demonstrated the efficiency of manifold learning methods in feature extraction and classification. Under the manifold assumption that the input data lie on or close to a low-dimensional data manifold, a series of state-of-the-art DR algorithms have been proposed and applied in HSI processing [28]–[30]. For example, Tenenbaum *et al.* [28] proposed isometric feature mapping (ISOMAP) which is a global geometric framework for searching a low-dimensional representation of the data that preserves the pairwise geodesic distances perfectly. To explore the locality of the data, Roweis and Saul [29] proposed locality linear embedding (LLE). Laplacian eigenmaps (LE) proposed by Belkin and Niyogi [30] is another classical transformation-based DR technique. In order to improve the robustness of these DR methods, Pu *et al.* [31] defined a novel spatial–spectral similarity metric named image patch distance (IPD) and developed IPD-based DR methods, including IPD-ISOMAP and IPD-LLE. Although these manifold learning methods can reveal data nonlinear structure, they often suffer from the so-called out-of-sample problem [32]. Therefore, some researchers introduced linear approximation techniques to improve them. The typical methods include locality preserving projection (LPP) [33], which is a linear version of LE, and neighborhood preserving embedding (NPE) [34], which linearly approximates to LLE.

In fact, most of the aforementioned methods can be unified into a common framework named graph embedding (GE) [35] that tends to compute a linear or nonlinear projection to map the high-dimensional input data to a desired low-dimensional submanifold space. In the GE framework, an adjacency graph is usually constructed to reveal the potential relationships between the data points, and a corresponding similarity matrix is utilized to code the graph structure. The traditional GE methods usually construct the adjacency graph by k-nearest neighborhood (kNN) or ϵ-ball neighborhood and compute the similarity between data points based on the pairwise Euclidean distance [36]. Since the performance of graph-based methods is demonstrated to be significantly determined by the quality of the predefined graph [37], researchers proposed numerous graph-construction methods to improve the characterization capability of the adjacency graph. In [36], an undirected weighted graph was constructed by l_1-based optimization to exploit the sparsity structure of data. Later, Ly *et al.* [38] proposed sparse graph-based discriminant analysis (SGDA) by using sparse representation to construct a block similarity matrix with class-specific labeled samples. Furthermore, Ly *et al.* [39] employed an l_2-based optimization to replace the l_1-based optimization in graph construction and developed collaborative graph-based discriminant analysis (CGDA), which greatly reduced the computational cost. To explore the potential properties of the data from other aspects, Li *et al.* [40] discussed a sparse and low-rank graph-based discriminant analysis (SLGDA), which attempts to capture essential data structure by constructing an informative graph with both sparsity and low-rank constraints. By combining the technique of kernel trick, Pan *et al.* [41] built a fast kernel sparse and low-rank graph and developed an extended version of SLGDA for HSI analysis.

Unfortunately, the preceding graph-based DR methods can only deal with the spectral vector-based (first-order) representation, which fails to consider spatial information. However, spatial information plays a critical role in HSI processing [42], [43]. On the one hand, some researchers suggested adding a segmentation preprocessing to capture the spatial information [44], [45] or using a sliding window to generate spatial–spectral features for graph construction [46]–[48]. On the other hand, the 3-D Gabor filter was introduced to extract multiscale spectral–spatial features for HSI classification [49]–[51]. Recently, to fully utilize spatial and spectral information simultaneously, the multilinear algebra (i.e., tensor theory) has been widely applied in HSI processing [52]–[55]. Zhang *et al.* [54] introduced a definition of spectral–spatial tensor representation of HSIs and proposed tensor discriminative locality alignment for HSIs spectral–spatial feature extraction. Later, a tensor LPP (TLPP) algorithm was introduced to the DR of HSIs [56]. The TLPP algorithm [53] is a tensorization of LPP that constructs adjacency graph with high-order tensor samples. To improve the robustness of TLPP, the graph construction is modified with the local region covariance descriptor by using the duality between feature space and original data space [57]. Following the idea of constructing more robust adjacency graph in tensor space to improve the performance of tensor graph-based methods, Pan *et al.* [58] proposed tensor SLGDA (TSLGDA). An *et al.* [59] proposed a patch tensor-based sparse and low-rank graph (PT-SLG) by using the tensor decomposition skillfully. Benefitting from the strong representation capability of spatial–spectral tensor features, these tensor graph-based methods provide promising improvement on performance.

However, the above-mentioned graph-based methods only construct a single graph to describe the intrinsic structure of data, which inevitably suffers from the problem of data structure being characterized one-sidedly. To address this problem, Zheng *et al.* [60] proposed a DR framework to perform feature extraction on the selected bands by defining the multiple Laplacian matrices. To represent the complex spatial–spectral structure of HSIs, Luo *et al.* [61] proposed spatial–spectral hypergraph discriminant analysis (SSHGDA) by constructing four different graphs. Although these multigraph methods have been successfully applied to the DR of HSIs, they can only deal with pixelwise vector representation of HSIs, which fails to capture the natural 3-D data structure of HSIs. Moreover, traditional fusion methods of multiple graphs often suffer from the selection issue of the weighting parameters. To this end, this article proposes a patch tensor-based multigraph embedding (PTMGE) framework for HSI analysis. The proposed PTMGE framework first constructs three different types of subgraphs from different hierarchies, which can complement each other. Then, a novel parameterless fusion strategy is

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designed to integrate three subgraphs together for comprehensively characterizing the intrinsic structure of HSIs. Finally, an eigenvalue decomposition problem that aims to preserve the similarity exactly is employed to compute the optimal low-dimensional projection matrix.

The rest of this article is organized as follows. Section II overviews the GE framework, sparse graph-based technique, and TLPP. Section III describes the proposed PTMGE framework. The experimental results are shown in Section V. Finally, Section VI draws the conclusion.

II. Related Work

A. Graph Embedding Framework

Let $X \in \mathbb{R}^{N\times M}$ denote a data set with N data points $x_i \in \mathbb{R}^M$. Using GE for DR is to find a projection matrix $P \in \mathbb{R}^{M\times R}$ ($R \ll M$) that maps the high-dimensional data X into a desired low-dimensional representation $Y \in \mathbb{R}^{N\times R}$, where $Y = P^T X$.

Specifically, in the GE framework, an intrinsic graph is defined to characterize the similarity between the data points that are to be maintained in the low-dimensional space. The intrinsic graph $G = \{X, W\}$ is an undirected weighted graph that is constructed by the vertex set X, where $W \in \mathbb{R}^{N\times N}$ is a real symmetric matrix and w_{ij} measures the similarity between the vertices i and j. Meanwhile, a penalty graph $G^p = \{X, W^p\}$ can be constructed as a graph whose vertices X are the same as those of G, but whose edge weight matrix W^p corresponding to the similarity characteristics is to be suppressed in the dimension-reduced feature space. Then, the objective function of GE is formulated as

$$\begin{aligned} P^\star &= \arg\min_{P^T X L^p X^T P} \sum_{j\neq i} \left\| P^T x_i - P^T x_j \right\|^2 w_{ij} \\ &= \arg\min_{P^T X L^p X^T P} \operatorname{tr}(P^T X L X^T P) \end{aligned} \tag{1}$$

where $P^\star$ is the optimal solution of (1), L is the Laplacian matrix such that $L = D - W$, and D is a diagonal matrix with $D_{ii} = \sum_j w_{ij}$. L^p is a diagonal matrix for scale normalization or the Laplacian matrix of a penalty graph G^p. Similarly, $L^p = D^p - W^p$, and D^p is a diagonal matrix with $D^p_{ii} = \sum_j W^p_{ij}$. Solving the minimization problem (1) is equivalent to solve the following generalized eigenvalue decomposition problem:

$$X L X^T P = \Lambda X L^p X^T P \tag{2}$$

where Λ is a diagonal matrix with the generalized eigenvalues, and the solution $P^\star$ is composed by the generalized eigenvectors associated with the first R smallest nonzero generalized eigenvalues.

Remarkably, Yan *et al.* [35] note that the intrinsic graph is necessary and the penalty graph is optional for the DR problem. It implies that constructing an intrinsic graph with a reasonable similarity criterion is critical for the GE framework. The Gaussian similarities with Euclidean distance and local neighborhood relationships are two typical criteria. In recent years, many other reasonable criteria were proposed for graph construction.

B. Sparse Graph-Based Technique

In [36], sparsity preserving GE (SPGE) was proposed by employing sparse representation to construct the adjacency graph. Specifically, the sparse representation estimates the ith data point x_i via the l_1 optimization

$$\min_{\alpha_i} \|\alpha_i\|_1 \quad \text{s.t. } \|x_i - X_i \boldsymbol{\alpha}_i\|^2 \leq \theta \tag{3}$$

where $\boldsymbol{\alpha}_i$ is a sparse coefficient vector, X_i is the data matrix except x_i itself, and $\theta > 0$ is a small tolerance. Then, the affinity matrix is formulated as

$$W^s_{i,i'} = \begin{cases} 0, & i = i' \\ \alpha_{i,i'}, & i > i' \\ \alpha_{i,i-1}, & i < i'. \end{cases} \tag{4}$$

To preserve the sparse representation exactly, a related projection P that maps the data into a low-dimensional space is learned from the following objective function:

$$\arg\min_{P^T X L^p X^T P} \operatorname{tr}(P^T X L^s X^T P) \tag{5}$$

where $L^s = D^s - W^s$ denotes the Laplacian matrix of the intrinsic graph and $D^s = \sum_i W^s_{ij}$ is a diagonal matrix. If there is a penalty graph constructed, L^p may be the Laplacian matrix associated with the penalty matrix W^p. Subsequently, Ly *et al.* [38] proposed SGDA and BSGDA to make full use of prior knowledge. A sparse and low-rank graph is constructed to obtain a further improvement in the performance of feature exaction in [40].

C. Tensor Locality Preserving Projection

TLPP was originally proposed in [53] by extending LPP into tensor space and first utilized for HSI classification in [56], which aims to seek a set of transformation matrices that map data into a low-dimensional space with local intrinsic information being preserved. Specifically, for N given data points $\boldsymbol{\mathcal{X}}_1, \boldsymbol{\mathcal{X}}_2, \ldots, \boldsymbol{\mathcal{X}}_N \in \mathbb{R}^{m_1\times m_2\times\cdots\times m_n}$, TLPP first constructs an adjacency graph $\mathbb{G}$ with k-nearest neighbors or ε-radius ball. Accordingly, a weight matrix $\mathbf{W}$ that characterizes the similarity between the data points is computed. Generally, this similarity matrix is desired to be sparse. If the nodes $\boldsymbol{\mathcal{X}}_i$ and $\boldsymbol{\mathcal{X}}_j$ are neighbors, $\mathbf{W}_{ij}$ will be a nonzero value, which is often computed by the heat kernel function, i.e., $\mathbf{W}_{ij} = \exp\{-((\|\boldsymbol{\mathcal{X}}_i - \boldsymbol{\mathcal{X}}_j\|^2)/(2\sigma^2))\}$, where σ is a kernel width parameter. Otherwise, $\mathbf{W}_{ij} = 0$. Then, a minimizing objective function is built to calculate the optimal transformation matrices $\mathbf{U}_1, \mathbf{U}_2, \ldots, \mathbf{U}_n$, whose formulation is defined as [53], [56]

$$\min_{\mathbf{U}_i|_{i=1}^n} \sum_{i,j=1}^{n} \left\| \boldsymbol{\mathcal{X}}_i \times_1 \mathbf{U}_1 \cdots \times_n \mathbf{U}_n - \boldsymbol{\mathcal{X}}_j \times_1 \mathbf{U}_1 \cdots \times_n \mathbf{U}_n \right\|^2_F \mathbf{W}_{ij}. \tag{6}$$

The above-mentioned optimization problem can be solved by an iteration algorithm. By fixing $\mathbf{U}_1, \ldots, \mathbf{U}_{r-1}, \mathbf{U}_{r+1}, \ldots, \mathbf{U}_n$, problem (6) can be rewritten as

$$\min_{\mathbf{U}_i} \sum_{i,j=1}^{n} \left\| \boldsymbol{\mathcal{Y}}^r_i \times_i \mathbf{U}_i - \boldsymbol{\mathcal{Y}}^r_j \times_r \mathbf{U}_r \right\|^2_F \mathbf{W}_{ij} \tag{7}$$

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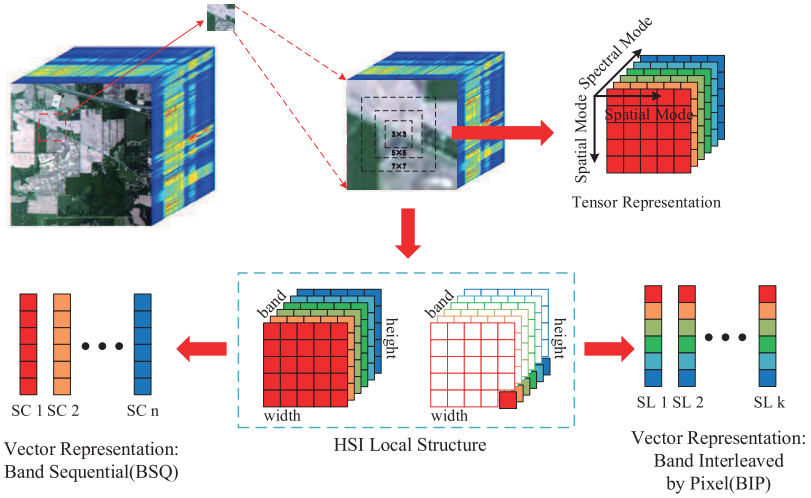


Fig. 1. Traditional spatial–spectral features and the tensor representation of the hyperspectral data cube.

where $\mathcal{Y}_i^r = \mathcal{X}_i \times_1 \mathbf{U}_1 \cdots \times_{r-1} \mathbf{U}_{r-1} \times_{r+1} \mathbf{U}_{r+1} \cdots \times_n \mathbf{U}_n$. After some simple derivations, problem (7) can be converted to the following generalized eigenvalue problem:

$$\left(\sum_{i,j} \left(\mathbf{Y}_i^{(r)} - \mathbf{Y}_j^{(r)}\right)\left(\mathbf{Y}_i^{(r)} - \mathbf{Y}_j^{(r)}\right)^T \mathbf{W}_{ij}\right)\mathbf{u} = \lambda\left(\sum_i \mathbf{Y}_i^{(r)}\mathbf{Y}_i^{(r)T} d_{ii}\right)\mathbf{u} \quad (8)$$

where $\mathbf{Y}_i^{(r)}$ is the matricization of $\mathcal{Y}_i^r$ and d_{ii} is the (i,i)th element of a diagonal matrix $\mathbf{D}$ whose entries are columnwise or rowwise sums of $\mathbf{W}$, i.e., $d_{ii} = \sum_i \mathbf{W}_{ij}$. By using the similarity measure defined in [31], an IPD-TLPP algorithm is derived for comparison in [57]. With the motivation of constructing a more robust intrinsic graph, the modified TLPP and TSLGDA algorithms are proposed. Later, PT-SLG is developed by computing the similarity matrix via patch tensor decomposition.

III. Proposed Method

HSIs appear as cubic data with two spatial modes and one spectral direction. The representation of the hyperspectral data can be simply divided into two categories: vector-based and tensor-based. For the vector-based representation of HSIs, there are two different cases. One is the spectral feature that is obtained by scanning the digital numbers in all bands into a vector; the other is traditional spatial–spectral feature that is generated by rearranging a local patch-based data cube to a first-order vector. There are two main approaches to conduct such a rearrangement: band interleaved by pixel (BIP) and band sequential (BSQ) [54]. The detailed illustration can be seen in Fig. 1.

From Fig. 1, it can be found that the traditional spatial–spectral feature and the spatial–spectral tensor representation are derived from the same typical local patch-based data cube. Especially, the spatial–spectral tensor has the same appearance with the typical local patch-based data cube. However, the tensor representation regards a local patch-based data cube as a whole without any reorganization, which can exactly represent the spatial–spectral structure of the hyperspectral data cube. For the traditional spatial–spectral feature, the BIP representation loses the spatially connected constraint among local pixels, while the BSQ representation fails to consider the spectrally connected constraint among spectral channels.

A. Patch Tensor-Based Multigraph Embedding Framework

As previously discussed, the traditional GE methods mainly focus on constructing single and multiple graphs with pixelwise spectral/feature vector representation, which generally fails to capture the spatial information and natural 3-D data structure of HSIs. Although the tensor-based GE methods overcome this shortcoming by using the tensor representation of HSIs for graph construction, they completely ignore the diversity among pixels from each patch tensor (or two neighboring patch tensors, as shown in Fig. 2) with the assumption that pixels in each patch tensor belong to the same class. To address these issues, a PTMGE framework is proposed, as shown in Fig. 3. In this framework, three types of subgraphs, namely patch tensor-based subgraph, pixel-based intrapatch subgraph, and pixel-based interpatch subgraph, are constructed and can complement each other. Among them, the first subgraph can make full use of the local 3-D spatial–spectral information of HSIs without any destruction and well characterize the holistic intrinsic structure of the spatial continuum in HSIs by tensor representation; the latter two subgraphs are constructed to describe the diversity among pixels from the same patch tensor and two neighboring patch tensors, respectively, which can extract more detailed and accurate information around the boundaries. The construction details of three kinds of subgraphs are presented as follows.

1) Patch Tensor-Based Adjacency Graph: The patch tensor-based adjacency graph is a weighted graph constructed by using the spectral–spatial tensor representation of HSIs. Let $\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_N\}$ denote the training set, where $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ is the ith tensor sample. To capture the intrinsic geometrical structure and simultaneously take the spatial information into consideration, a patch tensor-based

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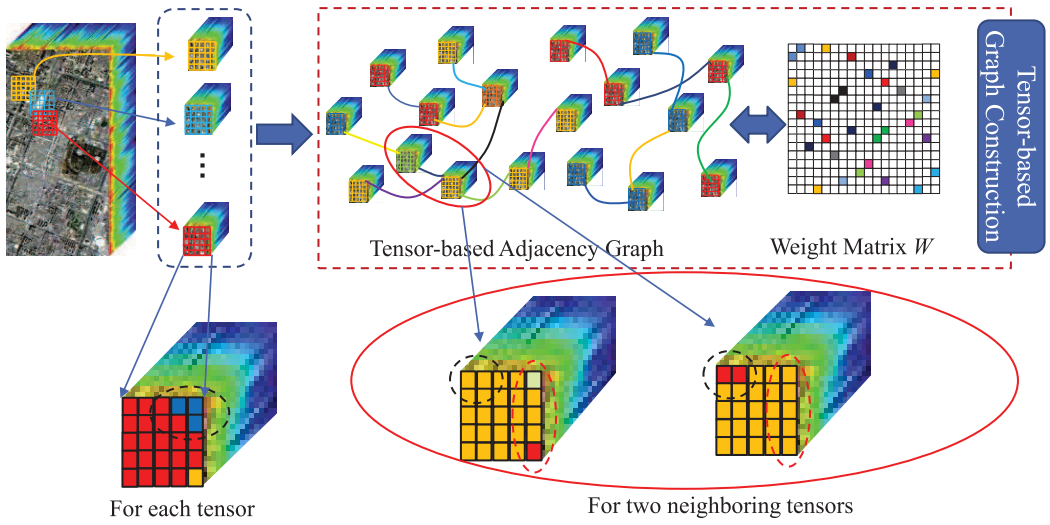


Fig. 2. Illustration of the diversity between pixels in patch tensors.

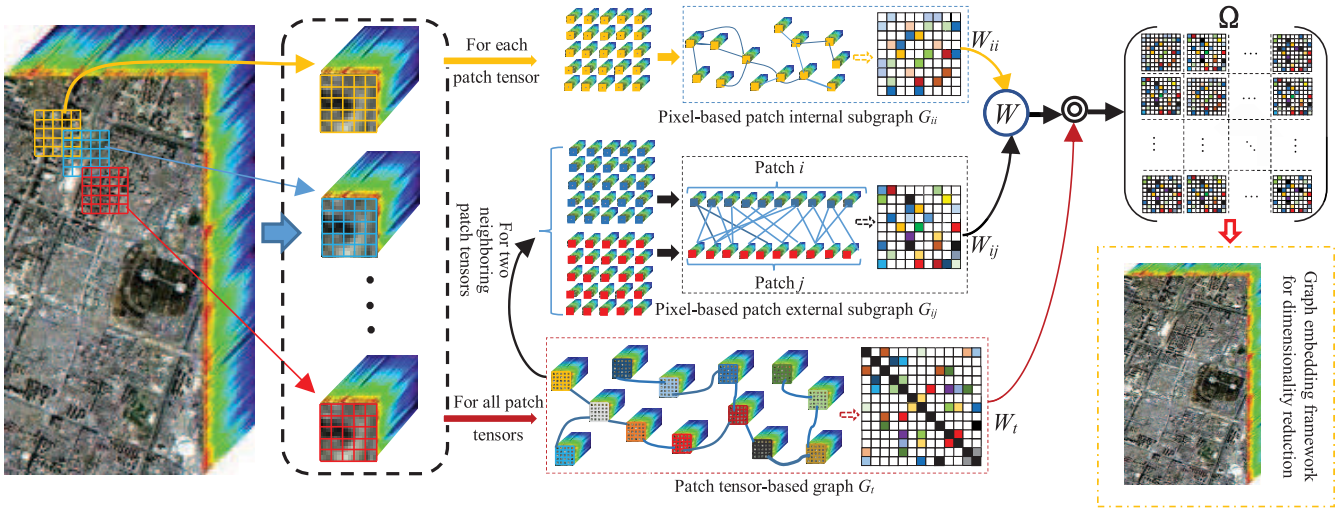


Fig. 3. Overall block diagram of the proposed method.

adjacency graph $G_t(\mathcal{X}, W_t)$ (as shown in Fig. 3) is constructed with the N tensor samples, where $W_t \in \mathbb{R}^{N\times N}$ is a related weight matrix. Specifically, $\text{diag}(W_t) = \mathbf{1}$, where $\mathbf{1}$ is a vector of all ones. Elementwise, W_t is defined as

$$W_t = \begin{bmatrix} 1 & W_t^{12} & \cdots & W_t^{1,N} \\ W_t^{21} & 1 & \cdots & W_t^{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ W_t^{N,1} & W_t^{N,2} & \cdots & 1 \end{bmatrix} \tag{9}$$

where W_t^{ij} represents the probability that data points $\mathcal{X}_i$ and $\mathcal{X}_j$ are belonging to the same class.

2) Pixel-Based Intrapatch Subgraph: The tensor-based adjacency graph regards all pixels in each local patch as a whole, which assumes that all pixels in a patch are strongly correlated and belong to the same class. However, the pixels in a local patch may be from different classes in practice. The tensor-based adjacency graph ignores the diversity between pixels in a local patch. In order to explore the intrastructure of the local patch, a pixel-based intrapatch subgraph (intra-P^{2}G, as shown in Fig. 3) is constructed for each patch. Simply, each intra-P^{2}G is built by the following procedures. For the ith local patch $\mathcal{X}_i = \{x_1, x_2, \ldots, x_l\}$, an adjacency graph G_{ii} is first constructed with the nodes $x_m \in \mathbb{R}^{I_3}$ $(m = 1, 2, \ldots, l;$ $l = I_1 I_2)$. Then, a corresponding sparse weight matrix $W_{ii} \in R^{l\times l}$ is calculated similar to the tensor-based adjacency graph. It should be remarked that $\text{diag}(W_{ii}) = \mathbf{0}$, where $\mathbf{0}$ is a vector with all zeros. Elementwise, the formulation of W_{ii} is

$$W_{ij} = \begin{bmatrix} W_{ij}^{11} & W_{ij}^{12} & \cdots & W_{ij}^{1,l^2} \\ W_{ij}^{21} & W_{ij}^{22} & \cdots & W_{ij}^{2,l^2} \\ \vdots & \vdots & \ddots & \vdots \\ W_{ij}^{l^2,1} & W_{ij}^{l^2,2} & \cdots & W_{ij}^{l^2,l^2} \end{bmatrix} \tag{10}$$

where W_{ii}^{mk} represents the similarity between pixels x_m and x_k in the ith patch tensor $\mathcal{X}_i$.

3) Pixel-Based Interpatch Subgraph: In the tensor-based graph, if two patch tensors are neighbors, the pixels from these two patches are forced to have the same similarity. However, there are some pixels from these two neighboring patches belonging to different classes in practice. For the purpose

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Algorithm 1: Summary of the PTMGE Algorithm

Input : Patch tensor dataset $\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_1, \cdots, \mathcal{X}_N\}$;
Step 1: Construct ptach tensor-based adjacency graph G_t with $\mathcal{X}$ by kNN algorithm;
Step 2: Compute the weight matrix W_t associated to G_t by (14);
Step 3: Construct a bipartite graph G_{ij} for the neighboring patch tensors $\mathcal{X}_i$ and $\mathcal{X}_j$ in G_t via k-NN algorithm, and calculate the corresponding similarity matrix W_{ij} by heat kernel function similar to (14);
Step 4: Construct an intrinsic graph G_{ii} for each patch tensor $\mathcal{X}_i \in \mathcal{X}$ with kNN algorithm, and compute the corresponding similarity matrix W_{ii} by heat kernel function similar to (14);
Step 5: Calculate the final weight matrix $\mathbf{\Omega}$ by (11);
Step 6: Compute the transiformation matrix P by solving (13);
Output: The projection matrix: P.

of exploring the diversity between pixels from two neighboring patches, a pixel-based interpatch subgraph (inter-P^{2}G, as shown in Fig. 3) is constructed for each two neighboring patches. Specifically, the inter-P^{2}G is a bipartite graph that is very different from the intra-P^{2}G and tensor-based adjacency graph. For two neighboring patches $\mathcal{X}_i = \{x_{i1}, x_{i2}, \ldots, x_{il}\}$ and $\mathcal{X}_j = \{x_{j1}, x_{j2}, \ldots, x_{jl}\}$, we construct the inter-P^{2}G G_{ij} by searching the neighbors of x_{im} from the patch $\mathcal{X}_j$ and searching the neighbors of x_{jk} from the patch $\mathcal{X}_i$. After the neighbor relationships are identified, a related sparse weight matrix $W_{ij} \in \mathbb{R}^{l\times l}$ is also computed. In addition, if there is no neighbor relationship between $\mathcal{X}_i$ and $\mathcal{X}_j$, the weight matrix W_{ij} is an $l \times l$ matrix of all zeros.

4) Subgraphs Fusion and Projection Learning: After these three different types of subgraphs are constructed, these subgraphs are fused into a final graph to implement DR. Thus, we design a novel way to construct the final affinity matrix $\mathbf{\Omega}$, whose formulation can be defined as

$$\mathbf{\Omega} = (W_t \otimes \mathbf{I}) * W \tag{11}$$

where $\otimes$ is the Kronecker product, $*$ denotes the Hadamard product, $\mathbf{I}$ is an $l \times l$ matrix of all ones, and W is a partitioned matrix consisting of the subblocks W_{ij} $(i, j = 1, 2, \ldots, N)$. In particular, W is defined as

$$W = \begin{bmatrix} W_{11} & W_{12} & \cdots & W_{1,N} \\ W_{21} & W_{22} & \cdots & W_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ W_{N,1} & W_{N,2} & \cdots & W_{N,N} \end{bmatrix}. \tag{12}$$

Since the HSI processing is a pixel-based task, the patch-based tensor data $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times I_3\times N}$ are unfolded along mode 3 to take the contribution of each pixel within the tensor training data into consideration. Based on the three-mode unfolding matrix $X \in \mathbb{R}^{I_3\times(I_1\times I_2\times N)}$, the GE framework is utilized to learn the optimal projection matrix $P^\star \in \mathbb{R}^{I_3\times R}$, where R is the reduced feature dimension. Therefore, the optimal projection matrix $P^\star$ can be obtained by solving the following eigenvalue decomposition problem:

$$XLX^T P = \Lambda P \tag{13}$$

where Laplacian matrix $L = D - \mathbf{\Omega}$, $D_{ii} = \sum_{j=1}^{I_1 I_2 N} \mathbf{\Omega}_{ij}$, and Λ is a diagonal eigenvalue matrix. The resulting project matrix $P^\star$ consists of the eigenvectors associated with the R smallest nonzero eigenvalues.

B. Summary of the PTMGE Algorithm

In the proposed PTMGE framework, there are three categories of subgraphs to be constructed. As mentioned earlier, many graph-construction approaches have been proposed, including kNN algorithm, ϵ-ball neighborhood, and l_1-based optimization. In particular, those subgraphs in the PTMGE can be built via the same method or different ways. This section summarizes the PTMGE that constructs subgraphs using the kNN algorithm.

Accordingly, the weight of the edge that connected the ith and jth nodes in the adjacency graph is calculated by the heat kernel function. Take the weight matrix W_t as an example, the element W_t^{ij} is defined as

$$W_t^{ij} = \exp\left\{-\frac{\|\mathcal{X}_i - \mathcal{X}_j\|^2}{2\sigma^2}\right\} \tag{14}$$

where σ denotes the kernel width parameter. Without loss of generality, all the weight matrices W_{ij} can be calculated in a similar way. Then, the final affinity matrix $\mathbf{\Omega}$ corresponding to the fused graph can be computed by (11). Finally, the desired transformation matrix P for DR can be obtained by solving the eigenvalue decomposition problem in (13). The complete PTMGE algorithm is summarized in Algorithm 1.

IV. Experimental Results and Analysis

A. Experimental Data Sets

To verify the performance of the proposed PTMGE, this section conducts a series of experiments on three popular HSI data sets [62]: Indian Pines, the University of Pavia, and Salinas.

The Indian Pines was acquired over an agricultural area in northern Indiana of USA in 1992. This data set was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor that generates 220 spectral bands from the wavelength range of 0.4–2.45 μm. In our experiments, a total of 200 bands after removing 20 water-absorption bands (bands 104–108, 150–163, and 220) are used. It covers a region of 145×145 pixels with 16 ground-truth classes. Table I lists the number of 3% training and 97% testing samples in each class. Specifically, if the number of training samples in any class is less than 3, we will randomly choose three training samples from the corresponding class.

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE INDIAN PINES DATA SET

Cl. #	Class name	training samples	testing samples
1	Alfalfa	3	44
2	Corn-notill	43	1385
3	Corn-mintill	25	805
4	Corn	8	229
5	Grass-pasture	15	468
6	Grass-trees	22	708
7	Grass-pasture-mowed	3	27
8	Hay-windrowed	15	463
9	Oats	3	19
10	Soybean-notill	30	942
11	Soybean-mintill	74	2381
12	Soybean-clean	18	575
13	Wheat	7	198
14	Woods	38	1227
15	Buildings-Grass-Trees-Drive	12	374
16	Stone-Steel-Towers	3	90
#	Total	319	9930

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE UNIVERSITY OF PAVIA DATA SET

Cl. #	Class name	training samples	testing samples
1	Asphalt	33	6598
2	Meadows	93	18556
3	Gravel	10	2089
4	Tree	15	3049
5	Painted metal sheets	7	1338
6	Bare Soil	25	5004
7	Bitumen	7	1323
8	Self-blocking bricks	18	3664
9	Shadows	5	942
#	Total	213	42563

The University of Pavia is a hyperspectral data set acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in Italy. After 12 noisy bands with spectral coverage from 0.43 to 0.86 μm are removed, a total of 103 bands remained in the image. It comprises 610×340 pixels with nine different land-cover classes. The number of 0.5% training and 99.5% testing samples is listed in Table II.

The last data set involved in the experiments is Salinas, which was also acquired by the AVIRIS sensor over the Valley of Salinas, Central Coast of California, in 1998. The image contains 512×217 pixels, with a spatial resolution of 3.7 m, and 16 different land-cover classes. It only preserves 204 spectral bands after removing 20 water-absorption bands. For this data set, a ratio of 0.3% is randomly chosen as training samples and the rest is used for testing. The number of training and testing samples from each class is listed in Table III.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE SALINAS DATA SET

Cl. #	Class name	training samples	testing samples
1	Brocoli-green-weeds-1	7	2002
2	Brocoli-green-weeds-2	12	3714
3	Fallow	6	1970
4	Fallow-rough-plow	5	1389
5	Fallow-smoth	9	2669
6	Stubble	12	3947
7	Celery	11	3568
8	Grapes-untrained	34	11237
9	Soil-vinyard-develop	19	6184
10	Corn-senesced-green-weeds	10	3268
11	Lettuce-romaine-4wk	4	1064
12	Lettuce-romaine-5wk	6	1921
13	Lettuce-romaine-6wk	3	913
14	Lettuce-romaine-7wk	4	1066
15	Vinyard-untrained	22	7246
16	Vinyard-vertical-trellis	6	1801
#	Total	170	53959

B. Experimental Setup

For the tensor-based methods, the selection of the spatial size of tensor samples is important. Since the spatial size not only depends on the consistency of different land-covers but also closely related to the number of ground-truth classes and the size of HSIs, there is still no common rule for determining the spatial size of tensor samples. Another problem is to determine how many features should be preserved. In practice, these two parameters are generally selected empirically and experimentally. Here, experiments on the three involved data sets are conducted to investigate the effects of the spatial size and the dimension of preserved features on the performance of the PTMGE. The window size is selected from the range of $\{3 \times 3,\ 5 \times 5,\ 7 \times 7,\ 9 \times 9,\ 11 \times 11\}$ for all considered three data sets. Fig. 4 shows the variation of overall accuracy versus different spatial window sizes and the number of preserved features. From Fig. 4, it is obvious that the overall accuracy is the highest at some window sizes and the proposed method provides stable classification accuracy with some dimensions. Specifically, 7×7 is the best spatial window size for the Indian Pines, and 9×9 is an appropriate size for the University of Pavia and the Salinas. Meanwhile, setting the number of preserved features as 30 is acceptable for these three data sets.

To evaluate the effectiveness of the proposed PTMGE, several state-of-the-art methods are used for comparison. Three vector-based GE methods, including LPP, SGDA, and SLGDA, are chosen as comparison methods. To make the comparison more impartial for vector-based methods, we add a preprocessing on LPP, SGDA, and SLGDA to take the spatial information into consideration, as in [48], and denote them as s-LPP, s-SGDA, and s-SLGDA, respectively. For these vector-based methods, the final dimension of reduced features impacts on the overall accuracy. However, considering the main purpose of DR, the number of preserved features is limited to be no larger than 50 in the experiments. In addition, three tensor-based GE methods, i.e., TLPP, IPD-TLPP, and PT-SLG, are used in the comparison experiments. In particular, IPD-TLPP is a reproduced method by adopting IPD defined in [31]. For both TLPP and IPD-TLPP, the spatial size of the tensor samples is set as 9×9, and the number of preserved features is set as same as that of PTMGE. The parameters involved in PT-SLG are set according to [59].

kNN is chosen as the classifier in our experiments. To eliminate the effects of randomly selecting training samples, all experiments are repeatedly run ten times.

C. Experimental Results

The experimental results of the Indian Pines data set are presented in Table IV and Fig. 5. Table IV lists the individual

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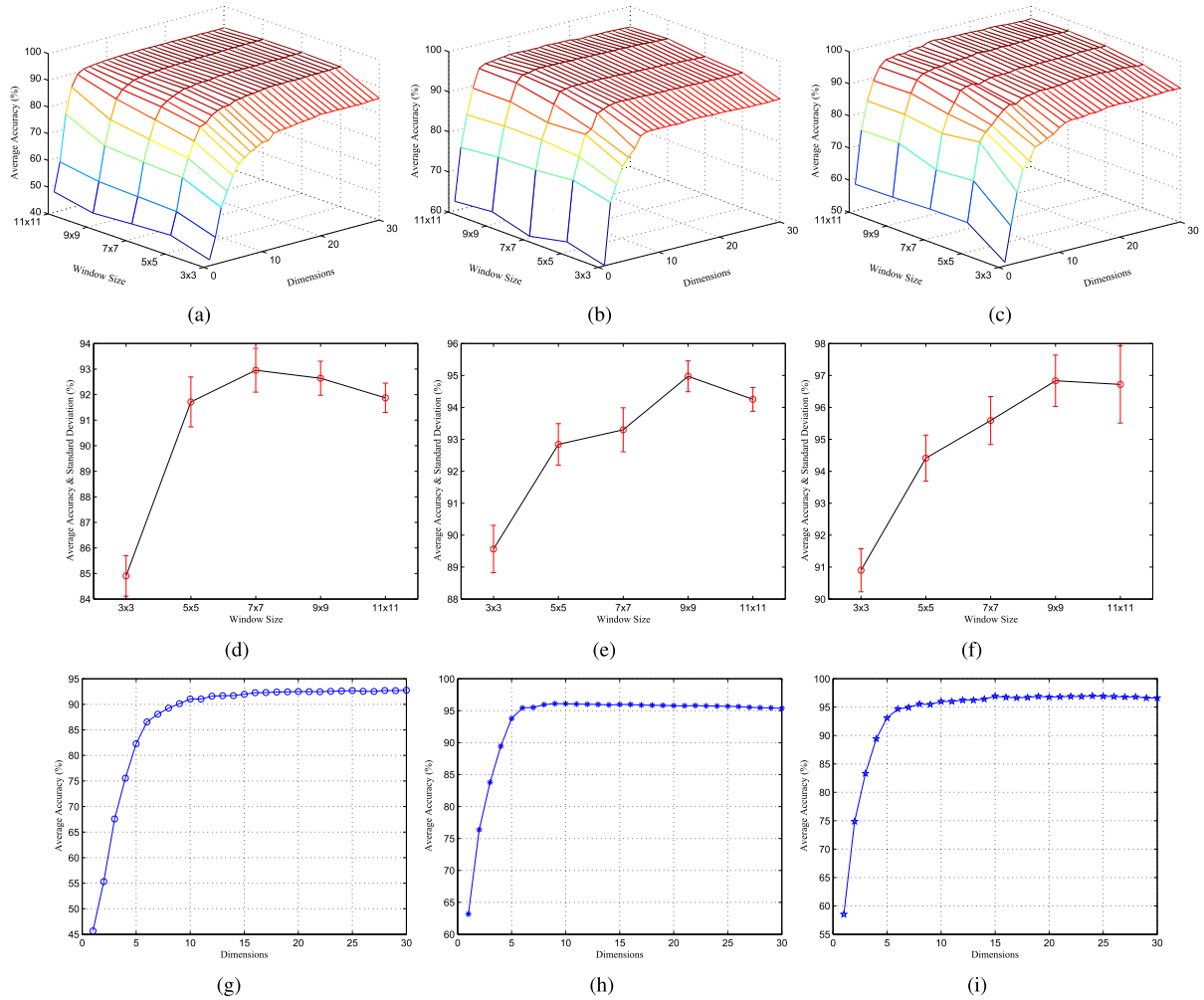


Fig. 4. (Top Row) Classification accuracy versus different local window size and different feature dimensions by using PTMGE. (a) Indian Pines. (b) Pavia University. (c) Salinas. (Middle Row) Classification accuracy (with standard deviations) versus different local window sizes when feature dimension is 30. (d) Indian Pines. (e) Pavia University. (f) Salinas. (Bottom Row) Classification accuracy versus different feature dimensions when the window sizes are 7×7, 9×9, and 9×9 for (g) Indian Pines, (h) Pavia University, and (i) Salinas, respectively.

TABLE IV

CLASSIFICATION ACCURACY (%) AND STANDARD DEVIATION FOR THE INDIAN PINES DATA SET (THE NUMBERS IN THE FIRST ROW ARE THE FINAL REDUCED DIMENSIONS)

Class	LPP (41)	s-LPP (35)	SGDA (50)	s-SGDA (32)	SLGDA (37)	s-SLGDA (28)	TLPP (30)	IPD-TLPP (30)	PT-SLG (30)	PTMGE (30)
1	33.95±5.42	86.98±6.34	27.21±13.24	36.28±19.49	10.93±6.16	89.30±13.77	85.17±8.91	**92.33±4.66**	42.12±16.58	91.40±5.10
2	37.83±6.01	70.71±3.09	44.80±9.17	73.39±6.04	50.61±6.35	83.42±3.62	78.06±2.08	80.75±1.98	85.89±2..86	**91.35±1.31**
3	27.70±5.22	66.67±4.20	36.20±7.86	68.66±6.09	36.87±7.61	76.36±5.94	76.32±3.31	79.70±4.04	83.23±3.01	**89.53±3.33**
4	21.13±4.71	57.17±8.54	10.74±7.41	40.52±8.11	9.52±3.73	75.57±13.49	70.59±7.74	70.52±6.27	63.30±10.35	**81.91±8.42**
5	58.55±7.59	75.16±3.06	75.35±5.77	89.15±4.76	69.42±4.20	64.09±19.15	79.17±6.01	86.95±1.62	**91.26±2.78**	86.52±1.85
6	82.70±3.17	86.21±4.84	89.60±4.35	96.02±4.01	89.58±3.64	96.43±3.88	94.39±1.80	**96.58±2.61**	93.53±3.43	94.72±3.60
7	72.00±12.65	98.40±3.20	9.54±4.40	34.40±27.78	10.80±6.71	99.60±1.20	97.78±9.47	98.80±1.83	**100±0.00**	94.40±7.63
8	80.09±8.44	90.11±4.86	96.55±3.55	97.69±5.14	93.60±3.53	91.40±7.56	96.98±1.79	97.69±2.05	96.40±3.05	**99.91±0.20**
9	31.18±8.34	71.76±11.10	18.24±8.12	59.41±18.27	2.88±2.35	90.00±20.22	93.31±1.85	97.06±4.74	63.16±28.73	**100±0.00**
10	36.24±4.33	71.78±4.38	28.06±11.31	60.58±9.21	43.91±3.07	86.68±5.59	83.67±3.42	87.31±4.05	79.96±3.83	**92.23±1.94**
11	57.10±3.12	81.75±2.69	83.64±3.32	86.98±4.18	58.49±3.27	84.12±2.39	88.64±1.74	89.58±2.09	86.28±2.37	**95.55±1.08**
12	22.33±2.46	61.20±2.66	8.87±4.80	47.44±10.75	34.83±4.43	86.38±5.99	67.70±3.40	74.66±3.44	68.79±4.67	**91.51±2.61**
13	62.81±10.11	93.52±2.99	65.73±15.81	95.58±1.82	58.94±13.61	**98.89±0.74**	97.74±7.89	98.49±1.92	86.79±4.67	94.62±2.26
14	78.71±5.17	92.10±2.36	94.82±3.25	97.86±1.54	92.48±2.71	96.68±2.61	89.99±7.70	95.60±1.58	95.21±2.36	**97.91±1.57**
15	25.99±2.61	61.95±13.31	30.16±8.51	80.75±6.65	36.71±7.16	89.95±5.48	81.47±5.27	84.41±10.22	74.16±6.25	**93.53±5.50**
16	76.00±6.29	**87.89±11.82**	81.78±5.22	85.56±14.84	35.22±15.73	83.78±29.99	82.56±6.48	81.00±10.91	83.68±13.98	80.00±11.03
OA	51.85±1.39	77.56±1.21	62.17±1.41	79.85±2.31	58.88±2.31	85.98±1.32	84.46±0.33	87.48±0.61	85.49±0.64	**93.37±0.67**
AA	50.27±1.23	78.34±2.05	49.76±1.62	71.89±4.12	45.90±1.92	87.04±1.71	85.22±1.32	88.21±0.81	80.86±2.96	**92.19±1.90**
κ	44.92±1.61	74.42±1.33	55.65±1.90	76.82±2.65	52.78±2.59	84.06±1.50	84.42±0.33	87.45±0.62	85.46±0.64	**93.36±0.67**

class accuracy, OA, average accuracy (AA), and kappa coefficient (κ) of the Indian Pines data set obtained by different methods. The classification maps in Fig. 5 show the visual results of Indian Pines data set using LPP, s-LPP, SGDA, s-SGDA, SLGDA, s-SLGDA, TLPP, IPD-TLPP, PT-SLG, and PTMGE. Obviously, for the Indian Pines data set, the proposed

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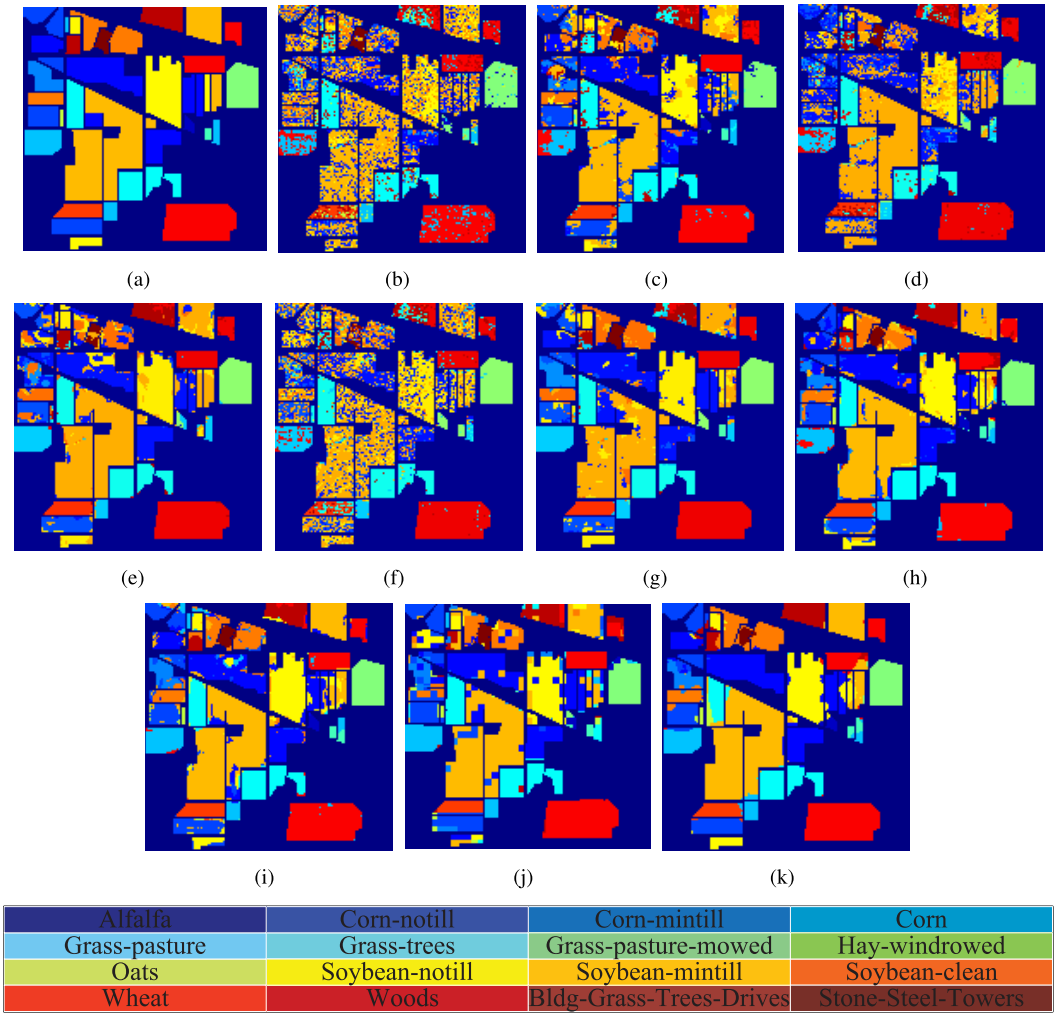


Fig. 5. Classification maps of Indian Pines by using different graph-based DR method. (a) Ground truth. (b) LPP. (c) s-LPP. (d) SGDA. (e) s-SGDA. (f) SLGDA. (g) s-SLGDA. (h) TLPP. (i) IPD-TLPP. (j) PT-SLG. (k) PTMGE.

TABLE V

CLASSIFICATION ACCURACY (%) AND STANDARD DEVIATION FOR THE PAVIA UNIVERSITY DATA SET (THE NUMBERS IN THE FIRST ROW DENOTE THE FINAL REDUCED DIMENSIONS)

Class	LPP (29)	s-LPP (26)	SGDA (49)	s-SGDA (38)	SLGDA (50)	s-SLGDA (33)	TLPP (30)	IPD-TLPP (30)	PT-SLG (30)	PTMGE (30)
1	48.13±3.84	71.70±7.20	82.96±5.80	92.38±1.92	54.05±5.29	84.95±2.94	83.64±5.33	85.52±4.79	81.05±4.30	**97.39±1.57**
2	86.58±2.68	90.02±2.18	98.22±1.31	98.12±1.53	84.02±3.25	93.08±1.53	97.80±0.63	98.70±0.86	96.46±1.52	**99.94±0.04**
3	33.66±9.32	63.35±9.75	33.55±11.00	72.05±8.01	54.53±7.62	76.79±4.28	66.75±6.82	67.56±5.02	66.82±4.60	**79.86±6.61**
4	60.16±6.67	67.76±7.98	69.20±12.24	80.41±6.01	83.84±3.95	**88.49±5.13**	86.28±2.68	85.89±3.92	75.45±8.56	86.59±2.62
5	99.01±0.37	99.65±0.51	97.55±0.73	99.22±0.32	99.45±0.10	11.14±9.43	99.54±1.18	99.87±0.33	89.98±6.61	**100±0.00**
6	31.73±4.17	61.25±4.72	17.78±6.24	45.26±10.93	46.79±5.20	85.96±3.04	80.41±4.67	**92.54±2.60**	80.11±5.03	84.72±3.44
7	39.04±6.91	61.66±11.53	20.26±15.49	52.00±10.93	37.30±6.32	91.01±5.42	76.66±3.27	76.73±7.02	70.50±8.58	**96.94±1.43**
8	57.96±8.94	69.33±3.63	73.76±10.51	83.38±5.20	47.35±8.62	75.66±5.11	73.06±4.26	78.69±3.74	65.71±3.87	**97.70±1.21**
9	43.59±17.20	68.33±10.56	**87.97±4.69**	82.14±7.56	41.54±10.99	76.72±9.91	76.80±6.65	65.11±7.98	55.97±11.68	85.46±6.71
OA	65.18±0.98	78.05±1.47	76.37±1.62	85.45±1.23	68.47±1.29	85.36±1.06	88.02±1.35	90.37±0.67	84.64±1.08	**95.21±0.76**
AA	55.54±2.70	72.56±1.93	64.59±4.38	78.33±1.59	60.99±2.33	76.00±3.17	82.33±1.25	83.40±1.89	75.78±2.68	**92.07±1.20**
κ	52.96±1.26	70.65±1.98	66.95±2.60	80.08±1.79	58.43±1.75	80.71±1.37	87.95±1.37	90.33±0.69	84.52±1.09	**95.20±0.76**

PTMGE achieves much better classification performance than other comparison methods. Besides the best performances of OA, AA, and κ obtained by the PTMGE, most of the highest class-specific accuracies are also achieved by the proposed PTMGE method. In particular, the proposed kNN-based PTMGE algorithm outperforms the other considered methods with the increase of more than 7% in OA, 17% in AA, and 7% in κ. It still provides competitive classification accuracy even if the number of training samples is very small. Taking classes 1 and 9 as examples, although there are only three training samples for them, the classification accuracies obtained by PTMGE achieve more than 90%, which are better than those of other comparison methods. Furthermore, the corresponding classification maps in Fig. 5 are consistent with these quantitative results. The proposed PTMGE provides much smoother classification regions than other considered

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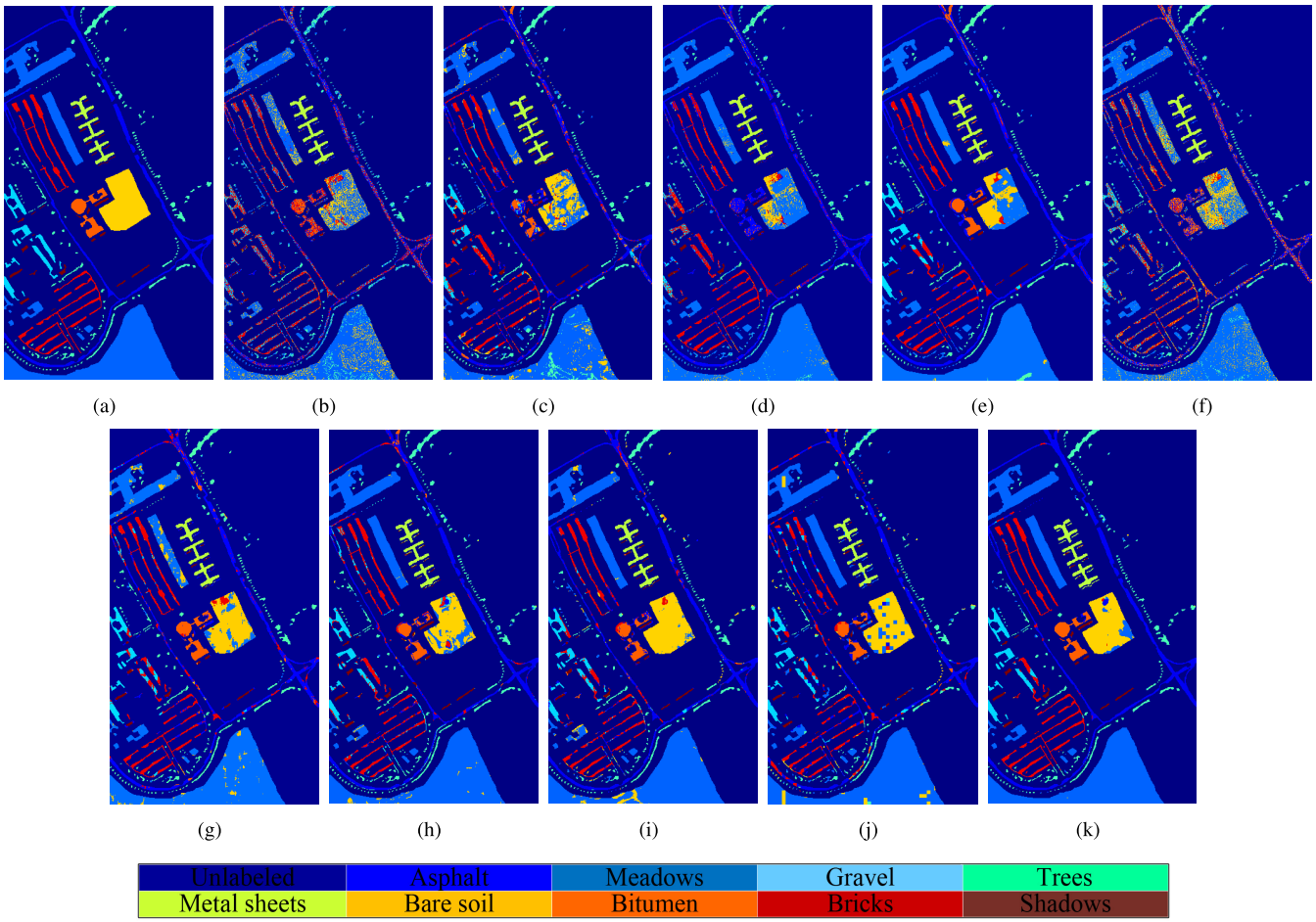


Fig. 6. kNN classification maps for the University of Pavia by using data transformed by different graph-based DR methods. (a) Ground truth. (b) LPP. (c) s-LPP. (d) SGDA. (e) s-SGDA. (f) SLGDA. (g) s-SLGDA. (h) TLPP. (i) IPD-TLPP. (j) PT-SLG. (k) PTMGE.

TABLE VI

CLASSIFICATION ACCURACY (%) AND STANDARD DEVIATION FOR THE SALINAS DATA SET (THE NUMBERS IN THE FIRST ROW DENOTE THE FINAL REDUCED DIMENSIONS)

Class	LPP (36)	s-LPP (31)	SGDA (50)	s-SGDA (47)	SLGDA (50)	s-SLGDA (39)	TLPP (30)	IPD-TLPP (30)	PT-SLG (30)	PTMGE (30)
1	76.89±14.19	85.29±9.67	94.77±1.56	96.12±1.79	94.92±2.82	97.03±1.43	95.41±1.93	94.30±3.32	94.69±1.88	**99.63±0.54**
2	91.17±3.90	92.10±7.20	98.29±0.50	99.74±0.08	98.51±0.48	99.59±0.26	88.90±7.04	91.67±5.12	95.62±1.99	**99.97±0.09**
3	26.92±7.58	61.83±10.81	64.24±8.39	30.08±9.75	56.81±10.15	47.98±22.14	79.73±8.72	93.18±10.48	87.09±11.29	**97.47±4.38**
4	32.79±4.45	68.86±10.32	94.35±1.95	86.88±8.47	93.49±5.36	91.24±7.39	87.79±6.99	84.91±10.67	89.49±3.82	**96.10±2.71**
5	57.46±11.06	79.24±6.04	96.05±1.24	88.06±20.89	96.69±1.93	95.98±1.79	96.77±2.14	94.71±3.47	86.91±3.55	**97.81±2.41**
6	96.81±3.30	98.09±3.13	97.75±1.70	**99.79±0.05**	98.65±0.68	99.61±0.35	99.01±1.39	98.97±1.30	96.42±1.96	99.63±0.32
7	81.45±7.74	85.03±8.48	99.15±0.23	98.76±0.70	98.94±0.59	98.94±0.50	82.25±7.03	90.41±7.76	95.76±1.69	**99.88±0.28**
8	60.97±3.13	70.53±4.47	92.78±4.17	91.08±6.37	91.18±5.05	85.82±2.45	84.90±3.38	85.02±4.22	80.40±4.60	**97.16±1.44**
9	82.97±6.55	97.55±1.47	97.15±1.18	97.72±2.25	95.79±2.14	97.45±2.24	97.76±2.23	97.63±3.35	96.34±2.38	**99.82±0.52**
10	40.69±7.91	66.32±11.04	78.33±8.83	78.44±5.89	79.73±5.28	77.73±9.20	81.08±9.24	80.18±9.82	85.03±4.10	**96.63±3.26**
11	31.04±22.64	62.22±16.56	17.00±13.45	25.19±23.10	48.87±29.05	41.86±22.59	83.62±16.83	89.74±7.16	85.12±5.91	**99.39±1.68**
12	33.51±7.91	72.43±7.36	81.20±13.40	96.42±5.36	86.27±16.77	93.76±8.39	86.86±13.60	85.34±9.73	93.97±3.74	**99.55±0.73**
13	31.36±12.95	73.80±18.74	85.43±11.26	76.43±28.04	88.66±17.45	59.27±22.24	96.58±3.85	96.36±5.78	90.44±3.14	**97.46±2.51**
14	26.45±16.86	72.09±11.05	58.73±21.85	64.39±22.06	77.42±14.41	89.12±3.74	95.10±4.70	94.52±4.61	81.06±11.67	**97.09±4.85**
15	39.03±3.76	54.50±5.58	17.22±15.92	35.87±15.81	21.77±16.58	64.84±10.76	70.72±5.14	79.84±5.07	78.51±5.27	**85.12±5.36**
16	54.83±11.54	75.17±10.33	69.51±9.27	70.89±7.52	75.39±12.35	76.89±6.38	92.34±4.42	85.17±7.62	85.70±7.03	**99.83±0.22**
OA	60.86±2.51	76.44±1.51	79.18±1.93	80.52±3.39	80.62±1.54	85.08±2.51	86.93±1.17	88.92±2.07	87.76±1.40	**96.70±0.82**
AA	54.02±3.28	75.94±2.91	77.62±1.74	77.13±6.04	81.44±2.55	82.32±3.89	88.68±1.94	90.12±2.22	88.91±1.00	**97.66±0.60**
κ	56.19±2.83	73.66±1.71	76.48±2.48	78.05±3.93	78.20±1.79	83.28±2.85	86.87±1.18	88.88±2.09	87.70±1.41	**96.37±0.82**

methods, especially for the regions of class 2 (Corn-no till), class 10 (Soybeans-notill), and class 11 (Soybeans-mintill).

The accuracy assessment and classification maps for the University of Pavia data set are presented in Table V and Fig. 6, respectively. The individual class accuracy, OA, AA, and κ obtained by the considered methods are listed in Table V. It can be seen that the highest OA as well as AA and κ are achieved by the proposed PTMGE algorithm, and it also provides better results than other considered methods in most classes. Specifically, the proposed PTMGE method

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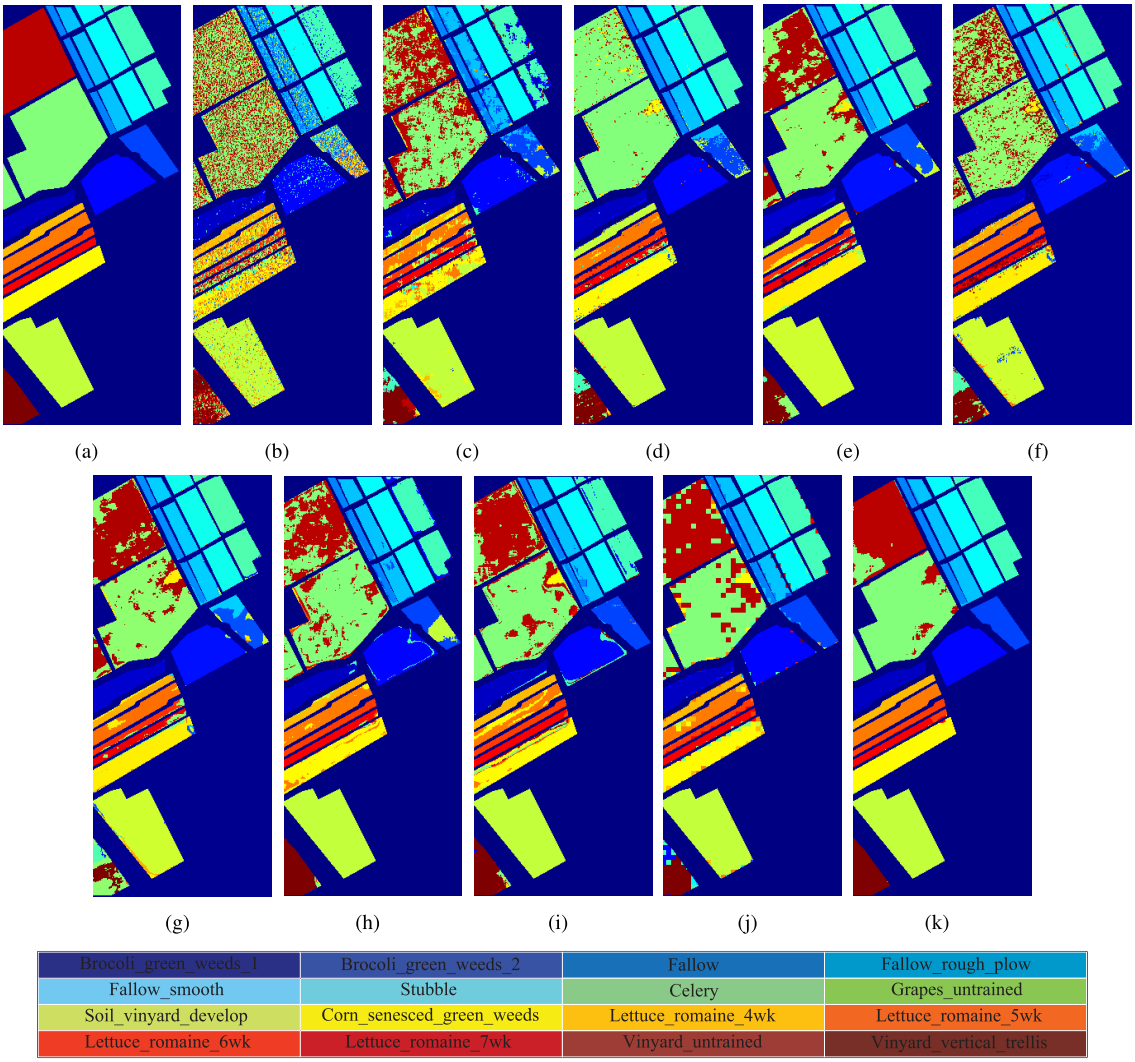


Fig. 7. kNN classification maps for Salinas by using data transformed by different graph-based DR methods. (a) Ground truth. (b) LPP. (c) s-LPP. (d) SGDA. (e) s-SGDA. (f) SLGDA. (g) s-SLGDA. (h) TLPP. (i) IPD-TLPP. (j) PT-SLG. (k) PTMGE.

outperforms the other comparison methods with the increase of about 5%–30% in OA, 8%–36% in AA, and 5%–42% in κ. Meanwhile, the classification maps in Fig. 6 show that the proposed PTMGE algorithm visually presents more smoother classification regions for most of the classes, especially for class 1 (Asphalt), class 2 (Meadows), and class 8 (Bricks), which are consistent with the results listed in Table V.

Table VI and Fig. 7 present the classification results of the Sanilas data set obtained by all related methods. According to the accuracy assessment listed in Table VI, it is easy to find that the performances of OA, AA, and κ offered by the PTMGE are also better than those of the methods such as LPP, s-LPP, SGDA, s-SGDA, SLGDA, s-SLGDA, TLPP, IPD-TLPP, and PT-SLG. Specifically, comparing with other considered methods, the PTMGE algorithm achieves an increase of more than 7.5% for OA, AA, and κ. Simultaneously, almost all of the highest individual class accuracies are achieved by the proposed PTMGE algorithm. Although some classes are highly similar, the PTMGE also provides promising classification performance. For example, classes 11, 12, 13, and 14 are Lettuce Romaine at different stages, but the highest classification accuracy of these classes is achieved by the PTMGE. Furthermore, the corresponding classification regions are smoother than those of other comparison methods in Fig. 7.

Overall, it can be found that the tensor graph-based methods (i.e., TLPP, IPD-TLPP, and PT-SLG) almost perform better than the vector graph-based methods (i.e., LPP, SGDA, and SLGDA). Moreover, although the spatial information is added on the vector graph-based methods, the tensor graph also outperforms them (i.e., s-LPP, s-SGDA, and s-SLGDA). In particular, comparing with LPP, TLPP achieves an increase of about 33% in OA, 35% in AA, and 39% in κ for the Indian Pines data set. Actually, similar conclusions can be obtained for the University of Pavia data set and the Salinas data set. These results illustrated that the tensor representation is more suitable for characterizing HSIs than the traditional spatial–spectral features. More importantly, it can be observed that the proposed PTMGE method obtains the best classification performance compared with the aforementioned single graph-based methods, including tensor graph-based methods and vector graph-based methods. This implies that the structure of

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hyperspectral data captured by the constructed multiple graphs is more robust than that described by a single graph. In addition, the PTMGE yields acceptable classification performance with a few training samples, which does not suffer from the problem of small training sample sizes.

V. Conclusion

In this article, we proposed a PTMGE framework for DR by constructing three different types of graphs to capture the intrinsic structure of HSIs. By constructing a tensor graph similar to TLPP, the spatial information and the local geometrical structure are captured for the DR of HSIs. For each pair of neighboring patch tensors, a bipartite graph is constructed for exploiting the detailed relationships between the patch tensors at the pixel level. For each patch tensor, a pixel-based subgraph is constructed to describe the diversity of pixels in individual patch tensors. By designing a novel multiple graphs fusion approach for projection learning, the structure described by different subgraphs is fully preserved in the low-dimensional space. Experiments with very small training sets are conducted on three real hyperspectral data sets to evaluate the proposed method, and the experimental results demonstrated the superiority of PTMGE over the state-of-the-art DR methods.

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Yang-Jun Deng received the B.Sc. degree in mathematics from Wuyi University, Jiangmen, China, in 2012, and the M.Sc. degree in applied mathematics from the University of South China, Hengyang, China, in 2015. He is currently pursuing the Ph.D. degree in signal and information processing with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu, China.

Since 2018, he has been studying in the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA, working as a Visiting Scholar with Prof. Q. Du to study on the dimensionality reduction of hyperspectral image. His research interests include pattern recognition and remote sensing image processing.

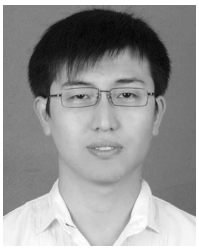


Heng-Chao Li (S'06–M'08–SM'14) received the B.Sc. and M.Sc. degrees from Southwest Jiaotong University, Chengdu, China, in 2001 and 2004, respectively, and the Ph.D. degree from the Graduate University of Chinese Academy of Sciences, Beijing, China, in 2008, all in information and communication engineering.

From 2013 to 2014, he was a Visiting Scholar along with Prof. W. J. Emery at the University of Colorado at Boulder, Boulder, CO, USA. He is currently a Professor with the Sichuan Provincial Key Laboratory of Information Coding and Transmission, Southwest Jiaotong University. His research interests include the statistical analysis of synthetic aperture radar images, remote sensing image processing, and signal processing in communications.

Dr. Li received several scholarships or awards, including the Special Grade of the Financial Support from the China Postdoctoral Science Foundation in 2009 and the New Century Excellent Talents in University from the Ministry of Education of China in 2011. He serves as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. He has been a Reviewer for several international journals and conferences, such as the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, the IEEE TRANSACTIONS ON IMAGE PROCESSING, *IET Radar, Sonar & Navigation*, *IET Signal Processing*, *IET Image Processing*, *Pattern Recognition*, the *International Journal of Remote Sensing*, *Remote Sensing*, and the *Canadian Journal of Remote Sensing*.

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Xin Song received the B.Sc. degree in information management and information system and the M.Sc. degree in economic management from Southwest Jiaotong University, Chengdu, China, in 2005 and 2011, respectively, where he is currently pursuing the Ph.D. degree in signal and information processing with the School of Information Science and Technology.

His research interests include hyperspectral image feature extraction and classification.



Yong-Jian Sun received the B.Sc. degree in clinical medicine from Taishan Medical College, Tai'an, China, in 1995, and the M.Sc. and Ph.D. degrees in orthopedics from Southern Medical University, Guangzhou, China, in 2002 and 2010, respectively.

He is currently a Professor with the Department of Traum Orthopedics, The Fifth Affiliated Hospital, Southern Medical University. His research interests include medical image processing and hyperspectral imaging.



Xiang-Rong Zhang (SM'14) received the B.Sc. and M.Sc. degrees in computer science and technology and the Ph.D. degree in pattern recognition and intelligent system from Xidian University, Xi'an, China, in 1999, 2003, and 2006, respectively.

She is currently a Professor with the School of Electronic Engineering, Xidian University. Her research interests include visual information analysis and understanding, pattern recognition, and machine learning.



Qian Du (S'98–M'00–SM'05–F'18) received the Ph.D. degree in electrical engineering from the University of Maryland at Baltimore, Baltimore, MD, USA, in 2000.

She is currently the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. She is also an Adjunct Professor with the College of Surveying and Geo-Informatics, Tongji University, Shanghai, China. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of the SPIE-International Society for Optics and Photonics. She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS). She has served as the Co-Chair for the Data Fusion Technical Committee of the IEEE GRSS from 2009 to 2013. She was the Chair with the Remote Sensing and Mapping Technical Committee of International Association for Pattern Recognition from 2010 to 2014. She was the General Chair for the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing, Shanghai, in 2012. She has served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JSTARS.

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